Classification Metrics

1. Recap: Why Accuracy Is Not Enough

Accuracy Formula

$$Accuracy = \frac{CorrectPredictions}{TotalPredictions}$$

Problem with Accuracy

Accuracy can mislead when dataset is **imbalanced**.

Example:

Airport Terrorist Detection

- 999 normal people
- 1 terrorist

If model predicts **everyone normal**, then:

- 999 correct
- 1 wrong

Accuracy = **99.9%**

But model failed to detect terrorist.

➞ So accuracy alone is dangerous.

2. Precision

Meaning:

Out of all cases predicted **Positive**, how many were actually Positive?

Formula:

$$Precision = \frac{TP}{TP + FP}$$

Where:

- TP = True Positive
- FP = False Positive

Use When:

False Positives are dangerous.

Example:

Spam Email Detection

If normal email marked as spam → important email missed.

So use **High Precision**

Easy Memory Trick:

Precision = Prediction Positive Check

3. Recall

Meaning:

Out of all actual Positive cases, how many model found correctly?

Formula:

$$Recall = \frac{TP}{TP + FN}$$

Where:

- FN = False Negative

Use When:

False Negatives are dangerous.

Example:

Cancer Detection

If patient has cancer but model says no cancer → dangerous.

So use **High Recall**

Easy Memory Trick:

Recall = Real Positives Found

4. Precision vs Recall

Metric	Focus
Precision	Correctness of Positive Predictions
Recall	Finding All Positives

5. False Positive vs False Negative

Error Type	Meaning
False Positive	Predicted Yes, actually No
False Negative	Predicted No, actually Yes

6. When to Use Which?

Situation	Important Metric
Spam Detection	Precision
Cancer Detection	Recall
Fraud Detection	Recall
Search Results	Precision
Terrorist Detection	Recall

7. F1 Score

Used when both Precision and Recall are important.

Formula:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Meaning:

Balanced score of Precision + Recall.

Why Harmonic Mean?

Punishes low values.

Example:

- Precision = 100%
- Recall = 50%

F1 becomes low.

So if one metric weak → F1 decreases.

8. Binary Classification

Two classes only:

- Yes / No
- Spam / Not Spam
- Disease / No Disease

Usually focus on Positive class.

9. Multi-Class Classification

More than 2 classes:

Example:

- Dog
- Cat
- Rabbit

Need Precision, Recall, F1 for each class separately.

Then combine results.

10. Averaging Methods in Multi-Class

Macro Average

Simple average of all classes.

$$\frac{Metric1 + Metric2 + Metric3}{3}$$

Use When:

All classes equally important.

Weighted Average

Average based on class size.

Large class gets more weight.

Use When:

Classes imbalanced.

Micro Average

Uses total TP, FP, FN globally.

Used for overall system performance.

11. Classification Report

Gives all metrics together:

- Precision
- Recall
- F1 Score
- Support

Support:

Number of actual samples in class.

12. Summary Table

Metric	Formula	Best Use
Accuracy	Correct / Total	Balanced Data
Precision	TP / (TP+FP)	Reduce FP
Recall	TP / (TP+FN)	Reduce FN
F1 Score	Combined	Need Both

13. Interview Questions

Q1 Why not use accuracy always?

Because imbalanced datasets can give fake high accuracy.

Q2 Precision vs Recall?

- Precision = Out of predicted positive, correct positives
- Recall = Out of actual positive, found positives

Q3 When use F1?

When both Precision and Recall matter.

14. Final

Precision:

How precise are positive predictions?

Recall:

How many positives recalled/found?

F1:

Balance of both